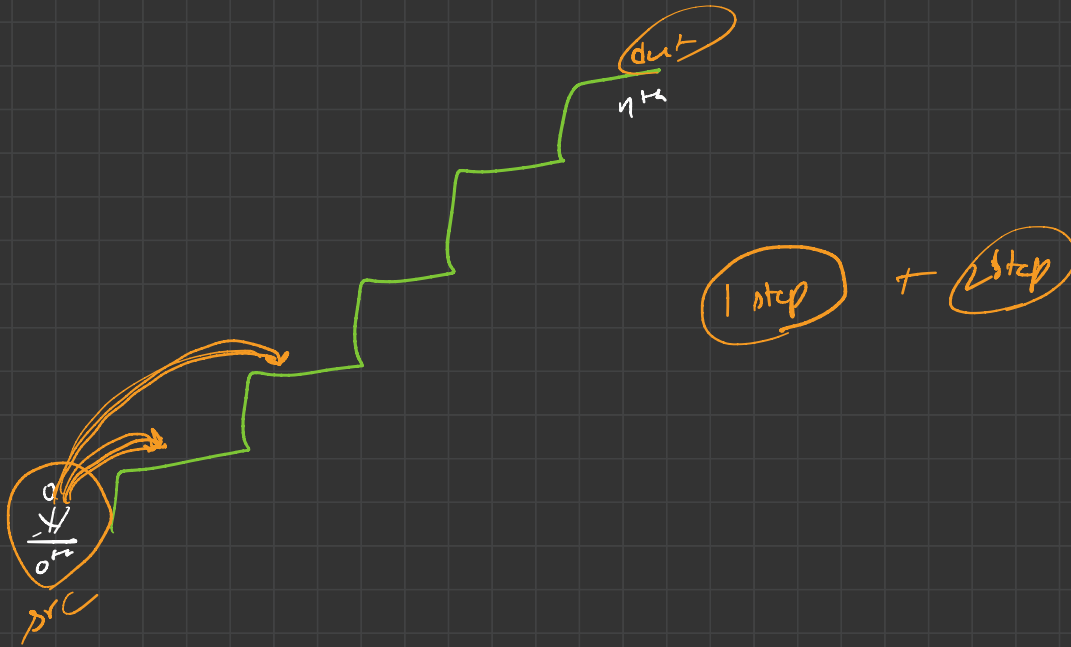
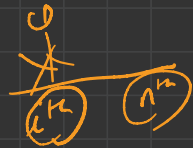
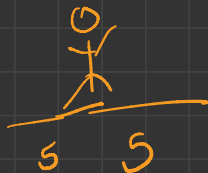
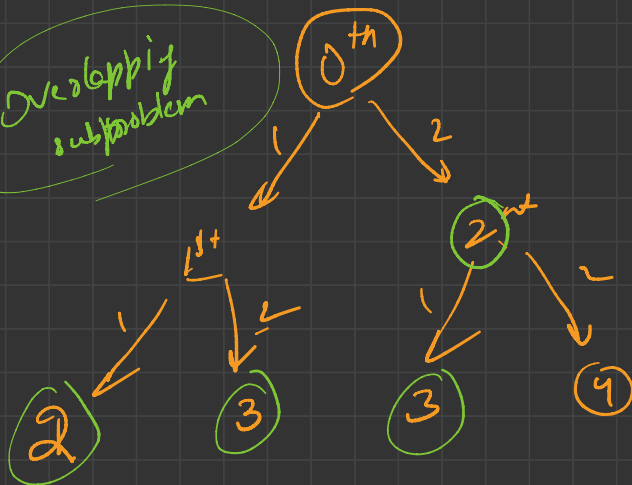



Dynamic Programming





overlapping
subproblem



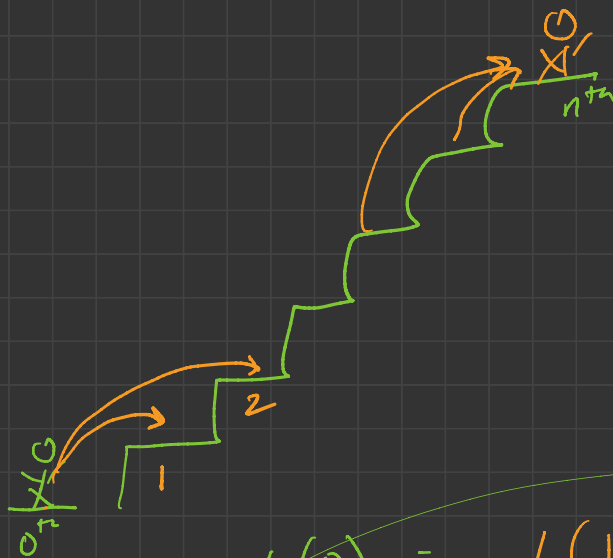
$$n-2 \quad n-1 \quad n$$

$$f(n) = f(n-1) + f(n-2)$$

$$f(n) = f(n+1) + f(n+2)$$

$$\underline{\underline{f(0)}} = \underline{\underline{f(1)}} + \underline{\underline{f(2)}} \rightarrow \text{pohuch yaf}$$

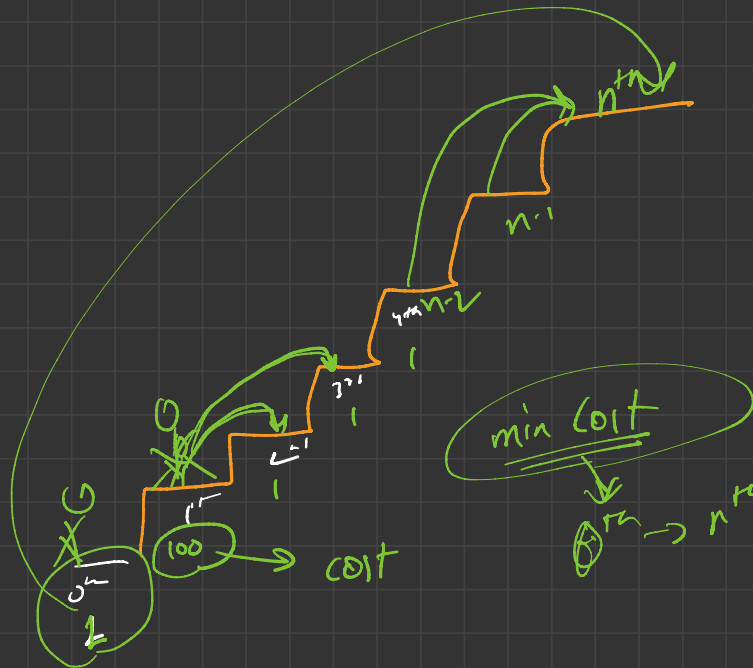




$$\underline{\underline{f(n) = f(n-1) + f(n-2)}}$$

$$\underline{\underline{f(0) = f(1) + f(2)}}$$

Min Cost Climbing Stair



① we can start from
0th or 1st stair

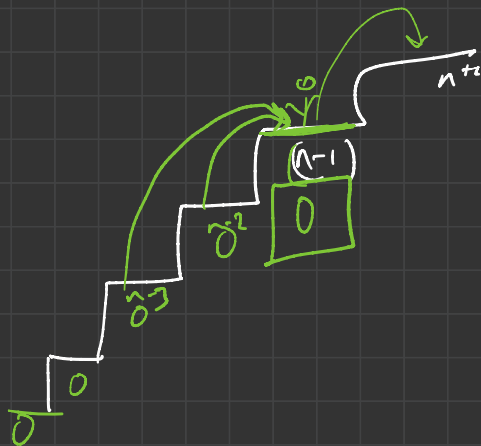
② we have to pay
cost, only then
we can move
1 or 2 steps
ahead

③ Return min cost
to reach top
floor

Recursion



$$f(n) = \min (f(n-1), f(n-2))$$



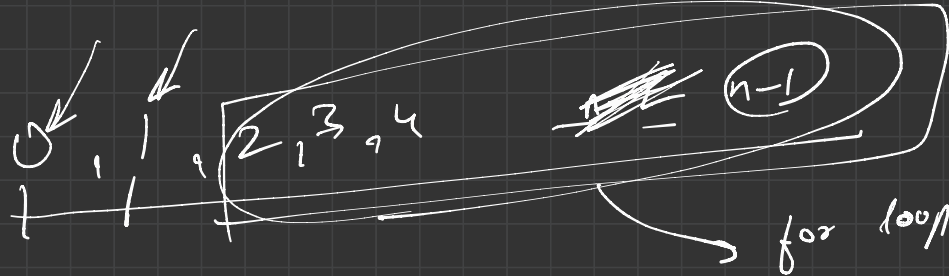
$$f(n-1) = \min (f(n-2), f(n-3)) + \text{cost}[i]$$

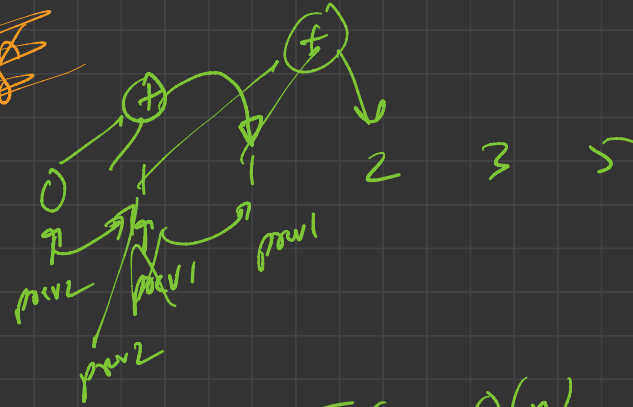
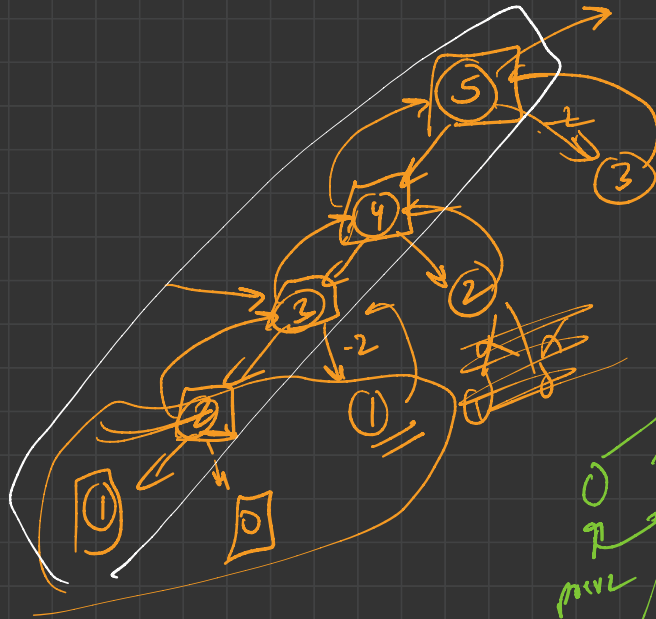
$$f(k) = \min \left(f(k-1), f(k-2) \right) + \text{cost}[k]$$

$\downarrow$
 $0 \rightarrow n-1$

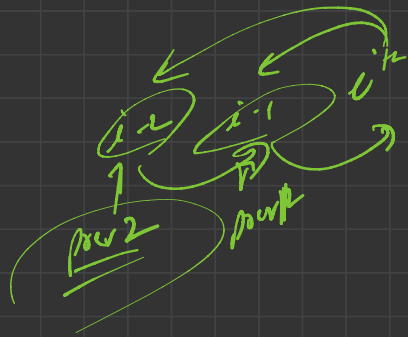
(i) Recursion + Memoization $\rightarrow$ $O(n)$
 $O(n) + O(n)$

(3) Tabulation $\rightarrow$ Bottom - up $\rightarrow$ T.C $\rightarrow O(n)$
 S.C $\rightarrow O(n)$ $\rightarrow$ $O(1)$





T.C $\rightarrow O(n)$
 S.C $\rightarrow O(1)$



→ Recursion + Memoization → T.C → $O(n)$ ✓
S.C → $O(n) + O(n)$ ✓

→ Tabulation → T.C → $O(n)$
S.C → $O(n)$

→ Space Optimisation → T.C → $O(n)$
S.C → $O(1)$ } →

